

HOWTO:Oracle 19c Enable TLS for FTPC11.X Installation

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Summary

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Question

FTPC11 Installs failed without the certification between DB and App server

Answer

The problem was reported for FactoryTalk Production Centre 11.01 but applies also to FactoryTalk Production Centre 11.X.

As all the connection in FTPC11 needs to be set as encrypted, this article is guide you for building the trusted relationship between app server and Oracle DB.

Oracle DB server

This section describes how to create self-signed wallet in Oracle DB, and export the certificate as a plain file.

1. mkdir -p C:\app\oracle\wallets
cd C:\app\oracle\wallets

For Linux:
mkdir /u01/app/oracle/wallets
cd /u01/app/oracle/wallets

2. orapki wallet create -wallet ./server_wallet -auto_login -pwd Rockwell@123

```
C:\app\oracle\wallets>orapki wallet create -wallet ./server_wallet -auto_login -pwd Rockwell@123
Oracle PKI Tool Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
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Operation is successfully completed.
```

Linux has the same command.

3. orapki wallet add -wallet ./server_wallet -dn "CN=<hostname>" -keysize 1024 -self_signed -validity 3650 -pwd Rockwell@123
(10 years validity)

```
C:\app\oracle\wallets>orapki wallet add -wallet ./server_wallet -dn "CN=win-14d4a9qrod7" -keysize 2048 -self_signed -val
idity 3650 -pwd Rockwell@123
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Version 19.3.0.0.0
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Operation is successfully completed.
```

Linux has the same command.

4. orapki wallet display -wallet ./server_wallet

Check if the certificate was created or not.

```
C:\app\oracle\wallets>orapki wallet display -wallet ./server_wallet
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Version 19.3.0.0.0
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Requested Certificates:
User Certificates:
Subject:      CN=win-l4d4a9qrod7
Trusted Certificates:
Subject:      CN=win-l4d4a9qrod7
```

Linux has the same command.

5. orapki wallet export -wallet ./server_wallet -dn "CN=<hostname>" -cert ./server_wallet/<hostname>.cert

```
C:\app\oracle\wallets>orapki wallet export -wallet ./server_wallet -dn "CN=win-l4d4a9qrod7" -cert ./server_wallet/win-l4d4a9qrod7.crt
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Version 19.3.0.0.0
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Operation is successfully completed.
```

Linux has the same command.

6. orapki wallet add -wallet C:\app\oracle\wallets\server_wallet -trusted_cert -cert C:\.ftpc-ca\certs\<hostname>.cert -pwd Rockwell@123

Import the certificate which we generated during the FTPC11 installation.

```
C:\app\oracle\wallets>orapki wallet add -wallet C:\app\oracle\wallets\server_wallet -trusted_cert -cert C:\.ftpc-ca\certs\win-l4d4a9qrod7.crt -pwd Rockwell@123
Oracle PKI Tool Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
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Operation is successfully completed.
```

For Linux: orapki wallet add -wallet ./server_wallet -trusted_cert -cert ./server_wallet/<hostname>.cert -pwd Rockwell@123

7. orapki wallet display -wallet ./server_wallet

Check if the client certificate was imported or not.

```
C:\app\oracle\wallets>orapki wallet display -wallet ./server_wallet
Oracle PKI Tool Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
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Requested Certificates:
User Certificates:
Subject:      CN=win-l4d4a9qrod7
Trusted Certificates:
Subject:      CN=win-l4d4a9qrod7,O=Rockwell Automation,ST=CA,C=US
Subject:      CN=win-l4d4a9qrod7
```

Linux has the same command.

Change the Listener file:

Open your Oracle DB directory, for example: C:\Installer\WINDOWS.X64_193000_db_home\network\admin\listener.ora, Linux should be /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora ; Make a copy before you change anything.

New value was highlighting the below sample content. Linux should be replaced by the path like /u01/app/oracle/wallets/server_wallet

```
SSL_CLIENT_AUTHENTICATION = FALSE
WALLET_LOCATION =
(SOURCE =
(METHOD = FILE)
(METHOD_DATA = (DIRECTORY = C:\app\oracle\wallets\server_wallet))
)
SID_LIST_LISTENER =
(SID_LIST =
(SID_DESC =
(SID_NAME = CLRExtProc)
(ORACLE_HOME = C:\Installer\WINDOWS.X64_193000_db_home)
(PROGRAM = extproc)
(ENVS = "EXTPROC_DLLS=ONLY:C:\Installer\WINDOWS.X64_193000_db_home\bin\oraclr19.dll")
)
)
LISTENER =
(DESCRIPTION_LIST =
(DESCRIPTION =
```

```
(ADDRESS = (PROTOCOL = TCP)(HOST = WIN-UV8BE9N4A3V)(PORT = 1521))
(ADDRESS = (PROTOCOL = TCPS)(HOST = WIN-UV8BE9N4A3V)(PORT = 2484))
(ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
)
)
```

Change the sqlnet.ora

Open your Oracle DB directory, for example: C:\Installer\WINDOWS.X64_193000_db_home\network\admin\sqlnet.ora, Linux should be /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/sqlnet.ora, Make a copy before you change anything.

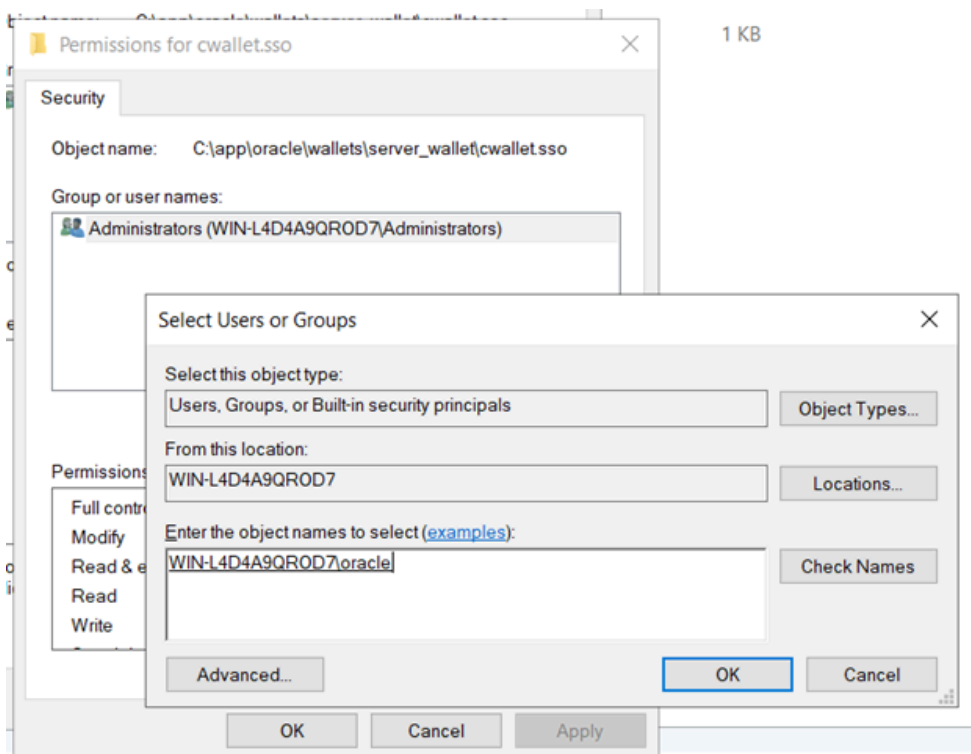
New value was highlighting the below sample content. Linux should be replaced by the path like /u01/app/oracle/wallets/server_wallet

```
SSL_CLIENT_AUTHENTICATION = FALSE
SQLNET.AUTHENTICATION_SERVICES=(TCPS,NTS)
NAMES DIRECTORY_PATH= (TNSNAMES, EZCONNECT)
WALLET_LOCATION =
(SOURCE =
(METHOD = FILE)
(METHOD_DATA = (DIRECTORY = C:\app\oracle\wallets\server_wallet))
)
```

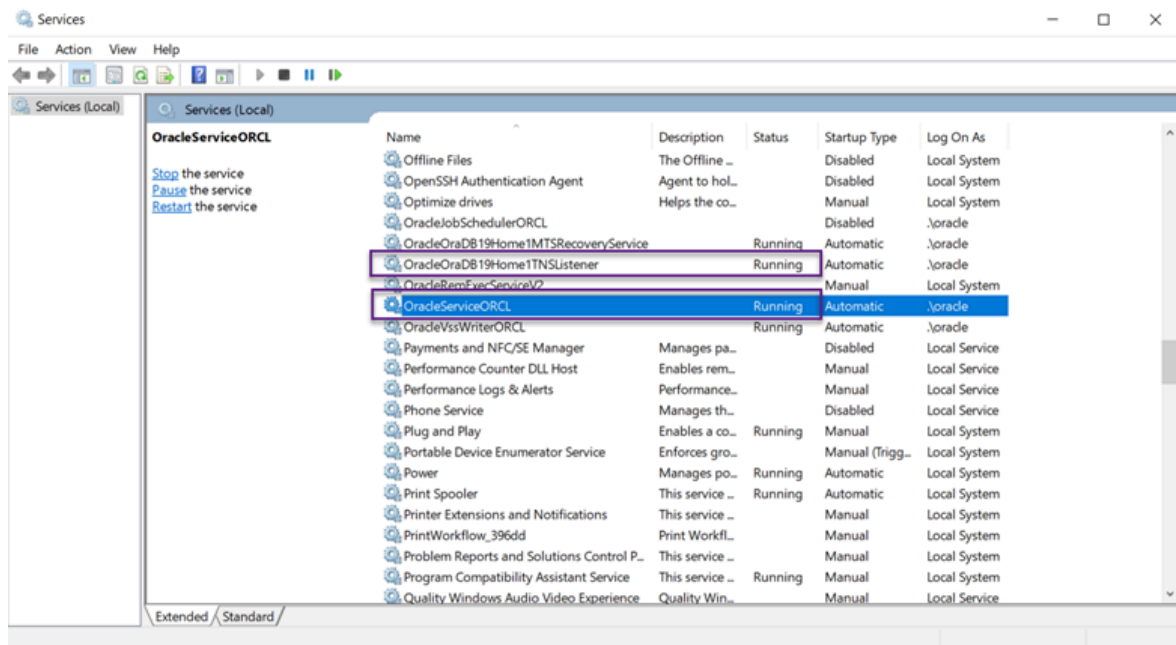
Change the permission of cwallet.sso

Open your wallet directory, for example: C:\app\oracle\wallets\server_wallet\cwallet.sso. Grant the user which you are using to start the oracle services.

Skip this part on Linux.



Restart Oracle Services and Listener



For Linux, you can restart listener using command "lsnrctl stop and start".

FTPC server (Can be regarded as Client)

Use the following example command to import the certificate exported from the database server into the trust store
 keytool -import -alias orakey -file C:\app\oracle\wallets\server_wallet\<hostname>.crt -keystore ftpc_truststore.p12.
 Linux has the same command, but make sure the file path is correct.

```
C:\Rockwell\ftpc-ca\certs>keytool -import -alias orakey -file C:\app\oracle\wallets\server_wallet\win-14d4a9qrod7.crt -k
eystore ftpc_truststore.p12
Enter keystore password:
Owner: CN=win-14d4a9qrod7
Issuer: CN=win-14d4a9qrod7
Serial number: 34f23a512fdd07fa8ff4643c214e26a2
Valid from: Wed Nov 22 13:27:27 CST 2023 until: Sat Nov 19 13:27:27 CST 2033
Certificate fingerprints:
  SHA1: F0:2F:B2:D8:F5:30:74:19:1E:11:F5:03:03:22:40:E2:16:78:F7:28
  SHA256: 17:EE:F3:EB:C7:2F:93:39:A3:9D:8D:DC:50:AA:E9:21:33:B4:27:A2:62:EF:3E:C8:F8:4F:C9:65:66:F2:96:93
Signature algorithm name: SHA256withRSA
Subject Public Key Algorithm: 2048-bit RSA key
Version: 1
Trust this certificate? [no]: yes
Certificate was added to keystore
```

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